

INVESTMENT RESEARCH

CONTENTS

**01 THE EVOLUTION OF
INVESTMENT
RESEARCH**

**02 WHERE AN EDGE
COMES FROM**

**03 READING LIST & TO
DO**

02 The Hypothesis Workshop. Challenge what you believe about markets, then find out who already tested it, and what they found.

WHERE WE LEFT OFF.

Three things were due for this session. The third one is the reason it exists.

01

ONE SOURCE PER CLAIM

Open access, read into your notes with read.

Bibliotheca/

02

YOUR STRATEGY, PUSHED

The objective and the benchmark entry.
Never Config/.env.

OBJECTIVE.md

03

ONE BELIEF

One paragraph on something you think is true about markets.

NOW WE TEST IT

THIS SESSION

CHALLENGE WHAT YOU BELIEVE ABOUT MARKETS.

Every belief you hold has already been
written down, tested and argued with.
Research starts by finding out by whom.

THE EVOLUTION OF INVESTMENT RESEARCH

Six acts · Prices → Value & risk → Efficiency → Anomalies → Discipline → Engineering

SIX ACTS.

Each act asks the question the one before could not answer
— dated by the works that forced it.

ACT I

PRICES

Can prices be predicted?

Prices look random; forecasters look like noise.

THE WORKS

1900 · Bachelier

1933 · Cowles

ACT II

VALUE & RISK

What is it worth, how much to hold?

Value is a model. Risk is priced.

THE WORKS

1934 · Graham & Dodd

1952 · Markowitz

1964 · Sharpe

ACT III

EFFICIENCY

Does anyone beat the market?

Proof is on you — and research is what efficiency pays for.

THE WORKS

1968 · Jensen

1970 · Fama

1976 · Ross

1980 · Grossman & Stiglitz

ACT IV

ANOMALIES

Why does the data misbehave?

Returns have structure. Mispricing has a mechanism.

THE WORKS

1993 · Fama & French

1993 · Jegadeesh & Titman

1979 · Kahneman & Tversky

1997 · Shleifer & Vishny

ACT V

DISCIPLINE

Do edges last?

Edges decay and the bar rises.

THE WORKS

2004 · Lo

2002 · Farmer

2013 · Asness, Moskowitz & Pedersen

2016 · Harvey, Liu & Zhu

ACT VI

ENGINEERING

How does an idea become a book?

The residual is isolated, scaled and learned.

THE WORKS

1995 · Grinold & Kahn

2021 · Paleologo

2020 · Gu, Kelly & Xiu

2022 · Guo, Wang, Ni & Shum

ACT I PRICES

Can prices be
predicted at all?

1900 Bachelier 1933 Cowles

ACT I · PRICES

DOW, BACHELIER & NELSON

Markets as signals.

THE WORK THAT CHANGED IT

1900

Théorie de la Spéculation
Bachelier

WHY IT MATTERS

Treated markets as an information system before formal theory existed.

KEY WORK

1900 Bachelier — Théorie de la Spéculation

1903 Nelson — The ABC of Stock Speculation

CORE IDEAS

- Prices embed collective information
- Trends reflect human behaviour
- Uncertainty can be modelled probabilistically

THE TAKEAWAY

The market is a barometer of all conditions.

WHERE THE EDGE IS

If prices carry information, so does their sequence. Trend, reversal and volatility clustering all start here.

**If prices carry
information... can
anyone read it better
than the crowd?**

ACT I · PRICES

COWLES, KENDALL & SAMUELSON

The first scoreboard.

THE WORK THAT CHANGED IT

1933

Can Stock Market Forecasters Forecast?

Cowles

WHY IT MATTERS

Put forecasters, Dow Theory and the price series themselves to a statistical test — and found little that beat chance. The random walk acquired evidence, then a proof.

KEY WORK

- 1933

Cowles — Can Stock Market Forecasters Forecast?
- 1953

Kendall — The Analysis of Economic Time-Series, Part I: Prices
- 1965

Samuelson — Proof That Properly Anticipated Prices Fluctuate Randomly

CORE IDEAS

- Track the call against the market, not against the story
- Price changes look random — a fact before it was a theory
- Properly anticipated prices must fluctuate randomly

THE TAKEAWAY

Put to the test, the forecasters were noise.

WHERE THE EDGE IS

Before you look for an edge, build the scoreboard. A strategy that is not measured against a passive alternative has not been tested at all.

ACT II VALUE & RISK

If prices wander... what is a security actually worth, and how much of it should you hold?

1934 Graham & Dodd 1952 Markowitz 1964 Sharpe

ACT II · VALUE & RISK

GRAHAM, DODD & DAMODARAN

Intrinsic value.

THE WORK THAT CHANGED IT

1934

Security Analysis

Graham & Dodd

WHY IT MATTERS

Turned investing from price-watching into a research discipline built on value and explicit assumptions. Williams gave value a formula — and Markowitz started from it.

KEY WORK

- 1934 Graham & Dodd — Security Analysis
- 1938 Williams — The Theory of Investment Value
- 1996 Damodaran — Investment Valuation

CORE IDEAS

- Value exists independently of price
- Valuation is a model, not a number
- Assumptions have to be written down

THE TAKEAWAY

Valuation is a bridge between stories and numbers.

WHERE THE EDGE IS

The gap between a story and a number is a hypothesis. Mispricing against intrinsic value is the oldest edge there is.

**If value exists.... how
do we allocate capital
across many bets?**

ACT II · VALUE & RISK

MARKOWITZ

Risk comes first.

THE WORK THAT CHANGED IT

1952

Portfolio Selection

Markowitz

WHY IT MATTERS

Formalised portfolio construction as a mathematical research problem.

KEY WORK

1952 Portfolio Selection

1959 Portfolio Selection: Efficient Diversification of Investments

CORE IDEAS

- Diversification is measurable
- Risk lives in covariance
- Portfolios beat individual assets

THE TAKEAWAY

Investment research starts with how you allocate capital.

WHERE THE EDGE IS

Edge lives in how you combine bets, not only in which ones you pick. Correlation is a return you can engineer.

**If risk can be
measured... which risk
deserves a reward?**

ACT II · VALUE & RISK

SHARPE & LINTNER

Pricing the risk.

THE WORK THAT CHANGED IT

1964

Capital Asset Prices

Sharpe

WHY IT MATTERS

Separated market compensation from research skill. Treynor and Mossin reached the same result independently.

KEY WORK

1964 Sharpe — Capital Asset Prices

1965 Lintner — The Valuation of Risk Assets

CORE IDEAS

- Beta earns a premium
- Performance attribution is born

THE TAKEAWAY

Not all risk is rewarded.

WHERE THE EDGE IS

Before you call it skill, ask whether it is beta in disguise. Strip market exposure first — most “edges” vanish.

ACT III EFFICIENCY

Does anyone actually beat the market — and if not, why would anyone try?

1968 Jensen 1970 Fama 1976 Ross 1980 Grossman & Stiglitz

ACT III · EFFICIENCY

JENSEN

Measuring skill.

THE WORK THAT CHANGED IT

1968

The Performance of Mutual Funds in the Period 1945–1964

Jensen

WHY IT MATTERS

Turned skill from intuition into a measurable residual: Jensen’s alpha, a regression intercept.

KEY WORK

1968 The Performance of Mutual Funds in the Period 1945–1964

2004 Berk & Green — Mutual Fund Flows and Performance in Rational Markets

CORE IDEAS

- Jensen’s alpha: performance after adjusting for risk
- Most active managers show no persistent skill
- Measurement precedes belief

THE TAKEAWAY

Investment research became a test, not a story.

WHERE THE EDGE IS

Set the bar before you look: the idea must beat a risk-adjusted benchmark, not zero. Measure, then believe.

**If skill is rare and fragile,
should it be assumed
— or must it be proven?**

ACT III · EFFICIENCY

FAMA

Markets as a discipline.

THE WORK THAT CHANGED IT

1970

Efficient Capital Markets

Fama

WHY IT MATTERS

Forced investment research to start from scepticism, not belief.

KEY WORK

1970 Efficient Capital Markets: A Review of Theory and Empirical Work

1991 Efficient Capital Markets: II

CORE IDEAS

- Prices reflect available information
- Skill is rare
- Evidence beats intuition

THE TAKEAWAY

Skill must be proven, not assumed.

WHERE THE EDGE IS

Assume the price already knows. Your idea must name the information it has not absorbed — or the reason it cannot.

If one beta cannot explain returns... how many risks are we being paid to carry?

ACT III · EFFICIENCY

ROSS

Risk is multi-dimensional.

THE WORK THAT CHANGED IT

1976

The Arbitrage Theory of Capital Asset Pricing
Ross

WHY IT MATTERS

Showed that returns must be driven by several independent sources of risk — even when we cannot yet name them.

KEY WORK

1976 The Arbitrage Theory of Capital Asset Pricing

CORE IDEAS

- No-arbitrage
- Multiple risk factors
- Theory before measurement

THE TAKEAWAY

Risk premia exist even when factors are unknown.

WHERE THE EDGE IS

Unnamed risk premia are candidate claims: a risk others refuse to hold may pay whoever holds it.

If nobody beats the market... why does anyone still pay for research?

ACT III · EFFICIENCY

GROSSMAN & STIGLITZ

The paradox that pays for research.

THE WORK THAT CHANGED IT

1980

On the Impossibility of Informationally Efficient Markets

Grossman & Stiglitz

WHY IT MATTERS

If prices reflected all information, nobody would be paid to gather it — so nobody would. Prices must stay imperfect enough to reward the research that improves them.

KEY WORK

1980 Grossman & Stiglitz — On the Impossibility of Informationally Efficient Markets

2015 Pedersen — Efficiently Inefficient

CORE IDEAS

- Information is costly; someone has to be paid to collect it
- Perfect efficiency is self-defeating
- Excess return is the wage of research, and the wage is competitive

THE TAKEAWAY

Efficient enough to be hard. Inefficient enough to be worth it.

WHERE THE EDGE IS

Your edge is the wage for information others found too expensive to gather. Ask what it costs to know this — and why the crowd will not pay it.

ACT IV ANOMALIES

**If markets are efficient...
why does the evidence
refuse to behave?**

1993 Fama & French 1993 Jegadeesh & Titman 1979 Kahneman & Tversky 1997 Shleifer & Vishny

ACT IV · ANOMALIES

FAMA, FRENCH & CARHART

Empirical reality.

THE WORK THAT CHANGED IT

1993

Common Risk Factors in Stocks and Bonds
Fama & French

WHY IT MATTERS

Showed that returns have structure, even in efficient markets.

KEY WORK

- 1992

Fama & French — The Cross-Section of Expected Stock Returns
- 1993

Fama & French — Common Risk Factors in the Returns on Stocks and Bonds
- 1997

Carhart — On Persistence in Mutual Fund Performance

CORE IDEAS

- Value, size, momentum
- Factor investing becomes scalable
- Research becomes systematic

THE TAKEAWAY

Markets are mostly efficient — but not simple.

WHERE THE EDGE IS

Known factors are the hurdle, not the idea. A new edge must survive after value, size and momentum are stripped out.

**If returns have structure...
which patterns survive
the cost of trading them?**

ACT IV · ANOMALIES

JEGADEESH & TITMAN · AMIHUD

Momentum meets liquidity.

THE WORK THAT CHANGED IT

1993

Returns to Buying Winners and Selling Losers

Jegadeesh & Titman

WHY IT MATTERS

Documented the anomaly that resists a risk explanation best — and priced the friction that decides who can actually trade it.

KEY WORK

- 1993

Jegadeesh & Titman — Returns to Buying Winners and Selling Losers
- 2002

Amihud — Illiquidity and Stock Returns: Cross-Section and Time-Series Effects

CORE IDEAS

- Past winners keep winning for three to twelve months
- Illiquidity is priced — and it is a cost you pay
- An anomaly is only as real as the trade that captures it

THE TAKEAWAY

The anomaly is free. The trade is not.

WHERE THE EDGE IS

Friction is edge. Illiquidity is priced, and the capacity to trade it decides who actually collects.

**If the anomalies are
real... who keeps
making the mistake?**

ACT IV · ANOMALIES

KAHNEMAN, TVERSKY, THALER & SHILLER

Human behaviour.

THE WORK THAT CHANGED IT

1979

Prospect Theory
Kahneman & Tversky

WHY IT MATTERS

Explained why inefficiencies survive — a direct challenge to rationality. Thaler took the biases to the data; Shiller showed prices move more than fundamentals can justify.

KEY WORK

- 1979 Kahneman & Tversky — Prospect Theory
- 1981 Shiller — Do Stock Prices Move Too Much?
- 1985 De Bondt & Thaler — Does the Stock Market Overreact?

CORE IDEAS

- Loss aversion and asymmetric risk preferences
- Overreaction shows up in prices — losers become winners
- Volatility exceeds what fundamentals can justify

THE TAKEAWAY

Risk is felt, not calculated.

WHERE THE EDGE IS

Systematic human error is the most durable mispricing. Ask which bias makes the counterparty sell to you — and keep doing it.

**If mistakes are
obvious... why
do they survive?**

ACT IV · ANOMALIES

SHLEIFER & VISHNY

Limits to arbitrage.

THE WORK THAT CHANGED IT

1997

The Limits of Arbitrage

Shleifer & Vishny

WHY IT MATTERS

Explained why being right is not enough: arbitrage is risky, costly and constrained. The same authors argued the value premium is a mistake, not a risk.

KEY WORK

- 1994

Lakonishok, Shleifer & Vishny — Contrarian Investment, Extrapolation, and Risk
- 1997

Shleifer & Vishny — The Limits of Arbitrage

CORE IDEAS

- Capital constraints and career risk
- The value premium as extrapolation error
- Anomalies persist where the arbitrageur cannot act

THE TAKEAWAY

Markets are stories with prices attached.

WHERE THE EDGE IS

Ask who cannot take the other side — constrained capital, career risk, forced sellers. Anomalies persist where they live.

ACT V DISCIPLINE

**So markets are rational...
and emotional. What survives
both — and for how long?**

2004 Lo 2002 Farmer 2013 Asness, Moskowitz & Pedersen 2016 Harvey, Liu & Zhu

ACT V · DISCIPLINE

LO

Markets adapt.

THE WORK THAT CHANGED IT

2004

The Adaptive Markets Hypothesis

Lo

WHY IT MATTERS

Rejected the random walk on the data in 1988, then reframed markets as evolving systems — reconciling efficiency with behaviour.

KEY WORK

- 1988Lo & MacKinlay — Stock Market Prices Do Not Follow Random Walks
- 2004Lo — The Adaptive Markets Hypothesis
- 2017Lo — Adaptive Markets: Financial Evolution at the Speed of Thought

CORE IDEAS

- Strategies have life cycles
- Edges are temporary
- Regimes matter

THE TAKEAWAY

Efficiency is not a state — it is a process.

WHERE THE EDGE IS

Every edge decays. Build in the signal that tells you the regime has changed — and be ready to retire the strategy.

**If markets adapt...
what exactly are
they adapting to?**

ACT V · DISCIPLINE

FARMER · ARTHUR · SOROS

Markets are ecologies.

THE WORK THAT CHANGED IT

2002

Market Force, Ecology and Evolution
Farmer

WHY IT MATTERS

Gave the Adaptive Markets Hypothesis its mechanism: strategies compete like species for a finite supply of mispricing, and the environment each one adapts to is mostly the others.

KEY WORK

1987	Soros — The Alchemy of Finance
1997	Arthur et al. — Asset Pricing under Endogenous Expectations in an Artificial Stock Market
2002	Farmer — Market Force, Ecology and Evolution
2011	Khandani & Lo — What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data

CORE IDEAS

- Strategies are species; mispricing is the food supply
- Prices feed back on the beliefs that set them
- Crowding is an ecological shock — August 2007 was one

THE TAKEAWAY

The environment a strategy adapts to is, mostly, the other strategies.

WHERE THE EDGE IS

Map the ecology before you enter it: who else runs this, with how much capital, on what horizon. An edge with no crowding measure is a species with no census.

If edges decay... what does durable research look like?

ACT V · DISCIPLINE

ASNESS

Research discipline in practice.

THE WORK THAT CHANGED IT

2013

Value and Momentum Everywhere

Asness, Moskowitz & Pedersen

WHY IT MATTERS

Proved that research survives markets through discipline.

KEY WORK

1997 The Interaction of Value and Momentum Strategies

2013 Asness, Moskowitz & Pedersen — Value and Momentum Everywhere

CORE IDEAS

- Factors persist but cycle
- Robustness beats intuition
- Behavioural foundations matter

THE TAKEAWAY

The best strategies do not look smart — they look resilient.

WHERE THE EDGE IS

Combine premia that fail at different times. Robustness across markets beats a perfect fit in one.

**If robustness beats intuition...
how many backtests
did it take to find yours?**

ACT V · DISCIPLINE

HARVEY, LIU & ZHU · MCLEAN & PONTIFF

The bar rises.

THE WORK THAT CHANGED IT

2016

... and the Cross-Section of Expected Returns

Harvey, Liu & Zhu

WHY IT MATTERS

Showed that after decades of searching the hurdle for a discovery must rise — and that anomalies shrink once published, because the market reads the papers too.

KEY WORK

2016 Harvey, Liu & Zhu — ... and the Cross-Section of Expected Returns

2016 McLean & Pontiff — Does Academic Research Destroy Stock Return Predictability?

2019 Arnott, Harvey & Markowitz — A Backtesting Protocol in the Era of Machine Learning

CORE IDEAS

- Hundreds of published factors, most of them noise
- A t-statistic of 2 is no longer evidence — aim for 3
- Anomaly returns roughly halve after publication
- How many were ever real is still contested — Jensen, Kelly & Pedersen, 2023

THE TAKEAWAY

The more you searched, the less your result means.

WHERE THE EDGE IS

Count your trials. An idea has to be justified before the backtest — the backtest can only fail to reject it.

ACT VI ENGINEERING

**If an edge is real... how does
it become a portfolio — and
what can a machine add?**

ACT VI · ENGINEERING

ROSENBERG · GRINOLD & KAHN

From theory to the desk.

THE WORK THAT CHANGED IT

1995

Active Portfolio Management
Grinold & Kahn

WHY IT MATTERS

Rosenberg turned multi-factor risk into a working model — BARRA — and Grinold & Kahn wrote the operating manual for running active research on top of it.

KEY WORK

- 1974

Rosenberg — Extra-Market Components of Covariance in Security Returns
- 1989

Grinold — The Fundamental Law of Active Management
- 1995

Grinold & Kahn — Active Portfolio Management

CORE IDEAS

- Risk models make the residual measurable, not just theoretical
- $IR \approx IC \times \sqrt{\text{breadth}}$ — skill times independent bets
- Many small bets beat a few big calls

THE TAKEAWAY

Skill is worth little without breadth.

WHERE THE EDGE IS

Edge scales with breadth. Ask how many independent bets a year your idea gives you — one call on the market is not a strategy.

**If everything has
exposure... what
is actually yours?**

ACT VI · ENGINEERING

PALEOLOGO

From ideas to portfolios.

THE WORK THAT CHANGED IT

2021

Advanced Portfolio Management

Paleologo

WHY IT MATTERS

Shows how investment research becomes real portfolios — by isolating the residual return and enforcing risk discipline.

KEY WORK

- 2021

Advanced Portfolio Management
- 2025

The Elements of Quantitative Investing

CORE IDEAS

- A return signal must be residualised from known risk factors
- Portfolio construction matters as much as idea generation

THE TAKEAWAY

Idiosyncratic return is what remains after risk is removed.

WHERE THE EDGE IS

Residualise. What is left after known factors are removed is the only part anyone will pay you for.

**If clean signals are rare...
who finds them faster:
humans or machines?**

ACT VI · ENGINEERING

GU, KELLY & XIU · DIXON, HALPERIN & BILOKON

Machine learning.

THE WORK THAT CHANGED IT

2020

Empirical Asset Pricing via Machine Learning

Gu, Kelly & Xiu

WHY IT MATTERS

Gu, Kelly & Xiu showed flexible models, disciplined out of sample, roughly double the predictive power of linear ones. Dixon, Halperin & Bilokon place the tools inside finance’s own tradition — not a black box.

KEY WORK

- 2020 Gu, Kelly & Xiu — Empirical Asset Pricing via Machine Learning
- 2020 Dixon, Halperin & Bilokon — Machine Learning in Finance: From Theory to Practice
- 2023 Kelly & Xiu — Financial Machine Learning

CORE IDEAS

- Nonlinearity and interactions carry real predictive power
- Learning replaces assumptions about the data-generating process
- Finance becomes a design problem, not only a modelling problem

THE TAKEAWAY

Investment research becomes a system that learns.

WHERE THE EDGE IS

Let the model learn structure you cannot write down — but only where you can say why it should exist.

If learning systems can model perception and decision together, how should we design investment research going forward?

ACT VI · ENGINEERING

GUO, WANG, NI & SHUM

Investment systems.

THE WORK THAT CHANGED IT

2022

Quant 4.0

Guo, Wang, Ni & Shum

WHY IT MATTERS

Research is no longer about finding isolated signals — it is about designing systems that continuously learn, adapt, and combine human structure with machine scale.

KEY WORK

2022 Quant 4.0: Engineering Quantitative Investment with Automated, Explainable and Knowledge-driven AI

CORE IDEAS

- Return signals live in representations, not raw data
- Machines explore scale; humans impose structure
- Research is a system, not a model

THE TAKEAWAY

Modern investing is engineered, not discovered.

WHERE THE EDGE IS

The durable edge is the system that keeps generating and retiring hypotheses — not any single one of them.

**INVESTMENT
RESEARCH DID
NOT EVOLVE BY
REPLACEMENT.**

**IT EVOLVED
BY ADDITION.**

THE ARGUMENT, DATED.

Five papers, thirty-four years. Read left to right and the case for research makes itself.

1970	1979	1980	1997	2004
FAMA	KAHNEMAN & TVERSKY	GROSSMAN & STIGLITZ	SHLEIFER & VISHNY	LO
THE CLAIM	THE COUNTERPARTY	WHY IT MUST EXIST	WHY IT PERSISTS	WHY IT KEEPS MOVING
Prices already reflect what is known. Beating the market is luck.	But people are not the calculators the claim assumes. Losing hurts more than winning feels good — and the mistake repeats.	If information costs money, prices cannot be perfect: somebody has to be paid to make them right.	Seeing the mispricing is not holding the trade. Capital, risk and horizon run out first.	Edges get crowded and decay, the environment shifts, new ones appear. Nothing stays put.
Efficient Capital Markets: A Review of Theory and Empirical Work	Prospect Theory: An Analysis of Decision under Risk	On the Impossibility of Informationally Efficient Markets	The Limits of Arbitrage	The Adaptive Markets Hypothesis

A return to research has to exist. It has a source. It persists. And it keeps moving.

THEREFORE

Research cannot be a one-time effort. The edge is in the process that finds the next one, not in any single signal.

THE EVIDENCE, DATED.

Five results, twelve years. Edges are real, they decay — and whether they were ever real is still being fought over.

2011	2016	2016	2016	2023
KHANDANI & LO	DANIEL & MOSKOWITZ	MCLEAN & PONTIFF	HARVEY, LIU & ZHU	JENSEN, KELLY & PEDERSEN
CROWDING	CYCLES	DECAY	FALSE POSITIVES	THE REBUTTAL
August 2007: quant books that shared a strategy unwound together in three days. An ecological shock, reconstructed.	Momentum crashes in sharp rebounds. A strategy works, gets crowded, breaks — and comes back.	Anomaly returns fall about 26% out of sample and about 58% once the paper is published.	Hundreds of factors published; most claimed findings are false once you correct for the number of tests.	Model the factors jointly and 82% replicate, in 13 themes across 93 countries.
What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data	Momentum Crashes	Does Academic Research Destroy Stock Return Predictability?	... and the Cross-Section of Expected Returns	Is There a Replication Crisis in Finance?

Real. Decaying. And still contested.

WHAT IT MEANS FOR US

Nobody will settle the factor fight for you. Deciding which edges are real, and for how long, is the job — and it never finishes.

NOTHING IS DISCARDED.

Where each stop on the two timelines sits once Lo’s frame is the umbrella.

INSIDE THE ADAPTIVE MARKETS HYPOTHESIS			Lo, 2004 — efficiency depends on the ecology
THEORY	WHO	ITS ROLE INSIDE THE FRAME	
EFFICIENT MARKETS	Fama, 1970	A special case: a stable environment, heavy competition and slow change.	
BEHAVIOURAL FINANCE	Kahneman, Tversky, Thaler	Where the maladaptive heuristics come from — shortcuts that worked in another environment.	
LIMITS TO ARBITRAGE	Shleifer & Vishny, 1997	The mechanism that lets inefficiencies last: capital, risk and horizon constraints.	
COSTLY INFORMATION	Grossman & Stiglitz, 1980	The equilibrium: markets cannot be perfectly efficient, or nobody would pay for information.	
EFFICIENTLY INEFFICIENT	Pedersen, 2015	The practical version: just enough inefficiency to pay the active managers who compete it away.	
MARKET ECOLOGY	Farmer · Santa Fe complexity, 1997–2002	The formal version: strategies as species, with interactions you can model.	
REFLEXIVITY	Soros, 1987	Feedback between participants and prices — part of what changes the environment.	

Each keeps its job. The frame is what lets them stop contradicting each other.

WHY RESEARCH GETS PAID.

01 IT HAS TO EXIST.

Grossman & Stiglitz, 1980 — if information is costly, someone is paid to gather it.

02 IT HAS A SOURCE.

Behavioural finance — heuristics that worked in another environment keep firing in this one.

03 IT PERSISTS.

Limits to arbitrage — capital, risk and horizon constraints on the people who see it.

04 IT KEEPS MOVING.

Adaptive markets — edges crowd and decay, the environment shifts, new ones appear.

SAID BEFORE ANYONE ELSE DOES

The frame is hard to falsify.

So make it operational: measure crowding, capacity, flows and the participant mix.

Grossman–Stiglitz is the stronger claim.

Use it for why the return to research cannot go to zero; use Lo for how it moves.

Berk & Green cut the other way.

‘The return exists’ is not ‘the client gets it’ — capital chases skill until the net return is gone. Say which claim you are making.

The edge is in the process, not in any single signal.

WHAT THIS BOOTCAMP BUILDS

A research process you can rerun on the next idea, not a fixed strategy.

ACT VII IS UNWRITTEN.

Four things the arc has not answered. Each one is where the next edge is most likely to come from.

Six acts are written. The seventh is the work you do in the next four sessions.

01

REGIMES YOU CAN SEE COMING

We name regimes confidently after the fact. Forecasting the turn in time to act on it is the difference between surviving a shift and explaining one.

WHO IS ON IT

Hamilton, 1989 · Ang & Timmermann, 2012 · the regime-switching and jump-model literature

WHERE IT LANDS HERE

Session 03: features that carry regime information.

02

A CENSUS OF THE CROWD

Who else holds this, with how much capital, on what horizon? That data is thin, late or proprietary — so crowding gets measured after it hurts.

WHO IS ON IT

Khandani & Lo, 2011 · flow and positioning research

WHERE IT LANDS HERE

Session 05: capacity, costs and position limits.

03

LEARNING THAT GENERALISES

Machine learning fits the past extremely well. The binding constraint is economic structure and sample size, not compute.

WHO IS ON IT

Gu, Kelly & Xiu, 2020 · Arnott, Harvey & Markowitz, 2019 · Jensen, Kelly & Pedersen, 2023

WHERE IT LANDS HERE

Session 04: the backtest protocol, written before the test.

04

RESEARCH THAT RUNS ITSELF

If edges decay, the asset is the pipeline: data, features, tests and attribution, re-runnable on tomorrow's idea. Automate the process, not the signal.

WHO IS ON IT

Guo, Wang, Ni & Shum, 2022 — Quant 4.0 · knowledge-driven AI

WHERE IT LANDS HERE

Every session: the strategy repository you started in session 01.

THE RESEARCH PROCESS IS DISCRETIONARY AT DESIGN AND SYSTEMATIC AT SCALE.

Scientific principles are what connect the
two.

WHERE AN EDGE COMES FROM

Every stop on the arc names a source of edge. Five of them survive scrutiny.

FIVE SOURCES OF EDGE.

An idea without a named source of edge is a pattern. Say which of these it draws on — and why that door is still open — before you test anything.

01**RISK PREMIA**

You are paid to hold a risk others refuse to bear.

ASK

What risk am I carrying — and who is paying me for it?

Sharpe · Ross · Fama & French

02**BEHAVIOURAL**

People err in the same direction, again and again.

ASK

Which bias makes the counterparty sell to me?

Kahneman & Tversky · Thaler · Shiller

03**LIMITS TO ARBITRAGE**

The people who know better cannot act on it.

ASK

Who cannot take the other side — and why not?

Shleifer & Vishny · Amihud

04**INFORMATION**

You measure something the price has not absorbed.

ASK

What do I see that the price does not — and why me?

Grossman & Stiglitz · Fama · Gu, Kelly & Xiu

05**IMPLEMENTATION**

Same signal — more of it kept after costs.

ASK

Can I capture a known edge better than the crowd does?

Markowitz · Grinold & Kahn · Paleologo

WHERE IDEAS COME FROM.

Six places to look. Each comes with the way it fools you.

01 PUBLISHED ANOMALIES

The factor zoo: hundreds of documented patterns, most with a paper behind them.

THE TRAP

Most are noise. Prefer ideas with a mechanism, replicated out of sample.

02 MARKET STRUCTURE

Index rebalances, option expiries, month-end flows, forced sellers.

THE TRAP

Crowded and small. Check capacity before you fall in love.

03 BEHAVIOURAL PATTERNS

Under-reaction to news, over-reaction to stories, the disposition effect.

THE TRAP

Name the bias and who holds it — or it is a story too.

04 FRICTIONS

Illiquidity, funding constraints, short-sale limits.

THE TRAP

The premium is real. The cost of collecting it is what you are testing.

05 YOUR OWN DATA

Something you can measure that others do not — or measure better.

THE TRAP

A new dataset is a hypothesis generator, not a hypothesis.

06 COMBINING KNOWN PREMIA

Value with momentum. Trend with liquidity. Carry with trend.

THE TRAP

Diversification is edge only if the pieces fail at different times.

SEVEN QUESTIONS BEFORE ANY BACKTEST.

01	Why does this edge exist?	GROSSMAN & STIGLITZ · FAMA
02	Who is on the other side, and why do they keep losing?	KAHNEMAN & TVERSKY · SHLEIFER & VISHNY
03	Is it a risk premium or a mispricing, and which one is the claim about?	ROSS · FAMA & FRENCH
04	How many independent bets a year does it give?	GRINOLD & KAHN
05	Does it survive costs, and how much capital can it hold?	AMIHUD · PALEOLOGO
06	When does it fail, and for how long?	LO · SHILLER
07	How many ideas were tried before this one?	HARVEY, LIU & ZHU · MCLEAN & PONTIFF

If the seven answers do not fit in one paragraph, it is not a hypothesis yet.

WHAT A GOOD IDEA LOOKS LIKE.

A belief is a sentence. A hypothesis names its edge, its counterparty, its test — and what would make you drop it.

THE CLAIM

One sentence, present tense, about a measurable relationship.

Liquid assets in an uptrend keep outperforming for weeks, not days.

THE SOURCE OF EDGE

Which of the five it draws on — and why that door should still be open.

Behavioural under-reaction, collected with an implementation edge.

THE OTHER SIDE

Who sells to you, and what keeps them doing it.

Cap-weighted holders who rebalance on size, not on trend.

THE TEST

The measurement that settles it, chosen before you look.

Rank by trend, weight by liquidity, hold one month, net of costs.

THE KILL SWITCH

The result that makes you abandon the idea instead of tuning it.

No excess return over the equal-weight control, net of costs, in two of three sub-periods.

WHICH ONE WOULD YOU ARGUE WITH?

Whose paper does your belief contradict — and are you comfortable with that?

If edges decay, what is the shelf life of the strategy you want to build?

What evidence would actually change your mind before you spend four months on it?

UP NEXT

NEXT: FEATURE ENGINEERING.

Session 03 is built on two things you bring:
the ten works, read — and one idea for a
strategy.

TO DO.

Two things, neither of them code. Session 03 opens by asking who did them.

01 READ THE TEN WORKS

The two timelines, in order — each answers the one before it. Four passes, thirty minutes each.

1970 Fama · Efficient Capital Markets

1979 Kahneman & Tversky · Prospect Theory

1980 Grossman & Stiglitz · On the Impossibility of Informationally Efficient Markets

1997 Shleifer & Vishny · The Limits of Arbitrage

2004 Lo · The Adaptive Markets Hypothesis

2011 Khandani & Lo · What Happened to the Quants in August 2007?

2016 Daniel & Moskowitz · Momentum Crashes

2016 McLean & Pontiff · Does Academic Research Destroy Stock Return Predictability?

2016 Harvey, Liu & Zhu · ... and the Cross-Section of Expected Returns

2023 Jensen, Kelly & Pedersen · Is There a Replication Crisis in Finance?

02 THINK ABOUT ONE IDEA

One paragraph for an investment strategy you would actually want to test — written the way section 02 asked. Not a model, not code: a belief.

-
- The claim: one sentence, present tense, measurable
 - Its source of edge: which of the five, and why it is still open
 - The other side: who sells to you, and what keeps them doing it
 - Which of the ten works argues against it — and what you would answer
 - The kill switch: the result that would end the idea

Write it for a claim in OBJECTIVE.md.

BACK TO THE PILLARS.

The coding, markets and math sessions behind today. Go back to one when you could not do its part with the AI off.

CODING · CODING FOUNDATIONS

- CF S04 The Agent Loop: Plan, Go, Review, Commit
the agent loop: plan, go, review, commit

MARKETS · FINANCIAL MARKETS

- ▣ FM S02 Prices, Total Returns and Time Value
CPI release dates
- ▣ FM S03 Rates, Bonds and the Yield Curve
duration, for the seeded case
- ▣ FM S08 Efficiency, Anomalies and Decay
efficiency, decay, fabricated citations; the history in six acts
- ▣ FM S11 Factor Models: Risk Factors and Return Signals
a risk factor or a return signal

MATH · MATHEMATICAL FINANCE

- ▣ MF S04 Estimation, Inference and Multiple Comparisons
t-statistics, p-values, multiple comparisons
- ▣ MF S06 Regression and Its Failures
reading a regression table

THE READING LIST.

Fifty-two works across six acts. The ten on the two timelines are the homework; the rest is the library you will build over the bootcamp.

ACT I & ACT II · PRICES → VALUE & RISK

DOW, BACHELIER & NELSON

- 1900 Théorie de la Spéculation
- 1903 The ABC of Stock Speculation

MARKOWITZ

- 1952 Portfolio Selection
- 1959 Portfolio Selection: Efficient Diversification of Investments

COWLES, KENDALL & SAMUELSON

- 1933 Can Stock Market Forecasters Forecast?
- 1953 The Analysis of Economic Time-Series, Part I: Prices
- 1965 Proof That Properly Anticipated Prices Fluctuate Randomly

SHARPE & LINTNER

- 1964 Capital Asset Prices
- 1965 The Valuation of Risk Assets

GRAHAM, DODD & DAMODARAN

- 1934 Security Analysis
- 1938 The Theory of Investment Value
- 1996 Investment Valuation

THE READING LIST.

Start with Fama (1970) and Grossman & Stiglitz (1980) if you only read two from here.

ACT III & ACT IV · EFFICIENCY → ANOMALIES

JENSEN · BERK & GREEN

- 1968 The Performance of Mutual Funds in the Period 1945–1964
- 2004 Mutual Fund Flows and Performance in Rational Markets

FAMA, FRENCH & CARHART

- 1992 The Cross-Section of Expected Stock Returns
- 1993 Common Risk Factors in the Returns on Stocks and Bonds
- 1997 On Persistence in Mutual Fund Performance

FAMA

- 1970 Efficient Capital Markets: A Review of Theory and Empirical Work
- 1991 Efficient Capital Markets: II

JEGADEESH & TITMAN · AMIHUD

- 1993 Returns to Buying Winners and Selling Losers
- 2002 Illiquidity and Stock Returns: Cross-Section and Time-Series Effects

ROSS

- 1976 The Arbitrage Theory of Capital Asset Pricing

KAHNEMAN, TVERSKY, THALER & SHILLER

- 1979 Prospect Theory
- 1981 Do Stock Prices Move Too Much?
- 1985 Does the Stock Market Overreact?

GROSSMAN & STIGLITZ

- 1980 On the Impossibility of Informationally Efficient Markets

LAKONISHOK · SHLEIFER & VISHNY

- 1994 Contrarian Investment, Extrapolation, and Risk
- 1997 The Limits of Arbitrage

THE READING LIST.

Start with Lo (2004) and Pedersen (2015) if you only read two from here.

ACT V & ACT VI · DISCIPLINE → ENGINEERING

LO

1988 Stock Market Prices Do Not Follow Random Walks

2004 The Adaptive Markets Hypothesis

2011 What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data

2017 Adaptive Markets: Financial Evolution at the Speed of Thought

ROSENBERG · GRINOLD & KAHN

1974 Extra-Market Components of Covariance in Security Returns

1989 The Fundamental Law of Active Management

1995 Active Portfolio Management

FARMER · ARTHUR · SOROS

1987 The Alchemy of Finance

1997 Asset Pricing under Endogenous Expectations in an Artificial Stock Market

2002 Market Force, Ecology and Evolution

PALEOLOGO

2021 Advanced Portfolio Management

2025 The Elements of Quantitative Investing

ASNESS · DANIEL & MOSKOWITZ

1997 The Interaction of Value and Momentum Strategies

2013 Value and Momentum Everywhere

2016 Momentum Crashes

GU, KELLY & XIU · DIXON · GUO

2020 Empirical Asset Pricing via Machine Learning

2020 Machine Learning in Finance: From Theory to Practice

2022 Quant 4.0

2023 Financial Machine Learning

THE FACTOR ZOO

2016 ... and the Cross-Section of Expected Returns

2016 Does Academic Research Destroy Stock Return Predictability?

2019 A Backtesting Protocol in the Era of Machine Learning

2023 Is There a Replication Crisis in Finance?

PEDERSEN

2015 Efficiently Inefficient: How Smart Money Invests and Market Prices Are Determined

HOW TO READ IT.

Fifty-two works is a library. The two timelines are the argument — start there.

One act a week. Six weeks, and the arc is yours.

START WITH THE TIMELINES · ALL TEN, IN ORDER — EACH ANSWERS THE ONE BEFORE IT

49 · The argument Fama 1970 · Kahneman & Tversky 1979 · Grossman & Stiglitz 1980 · Shleifer & Vishny 1997 · Lo 2004

50 · The evidence Khandani & Lo 2011 · Daniel & Moskowitz 2016 · McLean & Pontiff 2016 · Harvey, Liu & Zhu 2016 · Jensen, Kelly & Pedersen 2023

01

FOUR PASSES, THIRTY MINUTES

A paper is not a novel. Most of them give up their argument in half an hour.

- 01 Abstract, intro, conclusion — the question and the answer.
- 02 Tables and figures — the evidence, before the prose explains it to you.
- 03 Method — could you rebuild it with our data?
- 04 References — who they were arguing with. That is your next paper.

02

LEAVE A CARD BEHIND

Five lines per work, in your own notes. If you cannot write them, you have not read it.

THE CARD

Question · data and sample · result · why it might be wrong · what I would test.

WHERE

One markdown file per work, cited by year and full title.

03

BOOKS AND CLIPPINGS DIFFER

Not everything is read cover to cover, and clippings are noise with a date on them.

REFERENCE, BY CHAPTER

Graham & Dodd · Damodaran · Grinold & Kahn · Paleologo.

COVER TO COVER

Lo (2017) and Pedersen (2015) — the two timelines in book length.

NOW GO FIND OUT **WHO** **DISAGREES.**

Bring the ten works, read — and
one idea you are willing to see
killed.

QUESTIONS research@kaxanuk.mx

NEXT

SESSION 03

TOPIC

**FEATURE
ENGINEERING: THE
FEATURE LAB**

YOU LEAVE WITH

**A POINT-IN-TIME
UNIVERSE, ONE
FEATURE SCREENED
AND COUNTED.**

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